A (Very Naive) Study of the Intergalactic Medium Using the Lyman-α Forest

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1 Introduction

The intergalactic medium (IGM) contains the majority of baryonic matter in the Universe and plays a central role in structure formation and cosmic evolution. One of the most powerful observational probes of the IGM is the Lyman-α forest, which consists of a series of absorption features observed in the spectra of distant quasars.

As light from a high-redshift quasar propagates toward the observer, it is absorbed by intervening clouds of neutral hydrogen. Each absorber produces a Lyman-α transition at 1216 Å in its own rest frame, resulting in a forest of absorption lines blueward of the quasar's intrinsic Lyman-α emission line.

However, the interpretation of the Lyman-α forest is complicated by a number of systematic uncertainties such as modelling the intrinsic quasar continuum that is not directly observable in the forest region, contaminating metal absorption lines , Damped Lyman-α (DLA) systems introducing strong absorption and the quasar proximity effect modifying the ionization state of nearby gas.

In this work, I perform a simplified analysis of a quasar spectrum to estimate properties of the IGM, including the effective optical depth, column densities of absorbers, and the role of photoionization and scattering processes.

2 Data

I chose a high redshift (z=3.6), high SNR spectrum from eBOSS, with a clearly distinguishable forest. Furthermore, the forest has two regions A and B, one of which is contaminated by higher order Lyman series absorption lines, and needs to be treated more carefully. I only considered Region A for further analysis.

3 Analysis

3.1 Column Density Calculation

The observed flux density is defined as:

$$F_\lambda = \frac{dE}{d\lambda\, dA\, dt}. \tag{1}$$

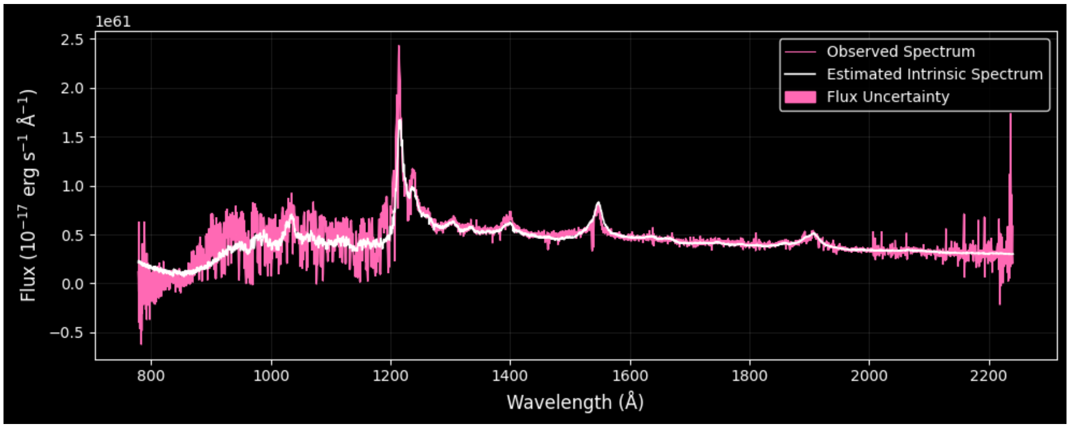


Figure 1: Raw spectrum from SDSS eBOSS

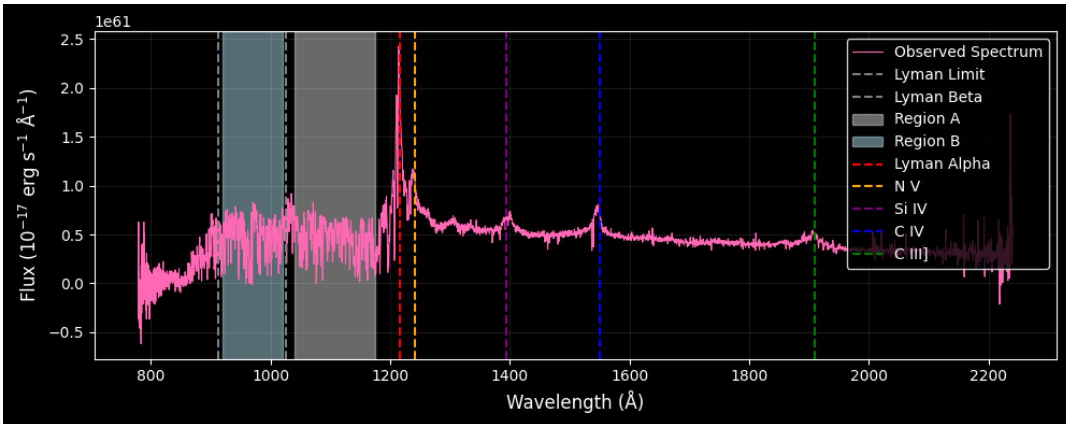


Figure 2: Spectrum with highlighted emission lines and forest regions

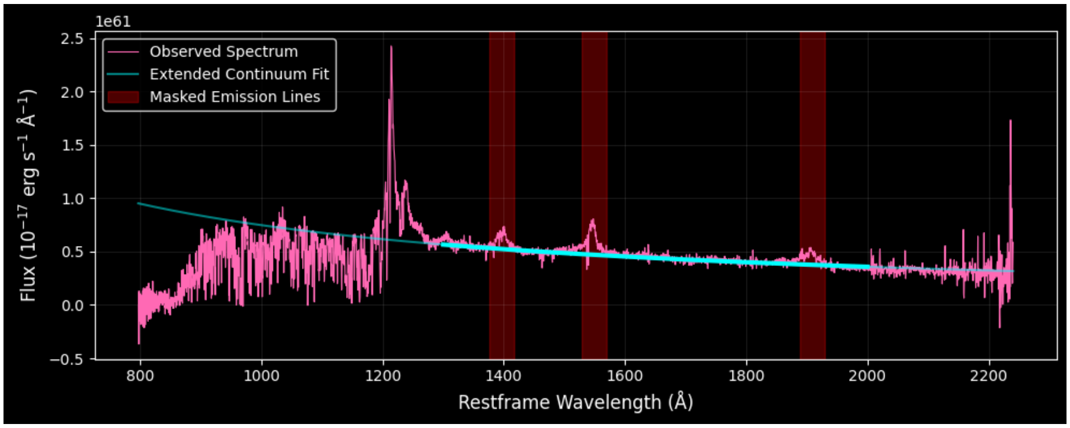


Figure 3: Continuum fit, masked emission lines and extension throughout the spectrum.

Due to cosmological redshift, the following transformations apply:

$$dE \rightarrow \frac{dE}{1+z}, \tag{2}$$

$$dt \rightarrow dt(1+z), \tag{3}$$

$$d\lambda \rightarrow d\lambda(1+z). \tag{4}$$

Substituting into the flux definition:

$$F_\lambda \rightarrow \frac{F_\lambda}{(1+z)^3}. \tag{5}$$

To obtain the luminosity,

$$L_\lambda = 4\pi d_C^2\, F_\lambda (1+z)^3 = 4\pi d_L^2\, F_\lambda(1+z) \tag{6}$$

A major challenge arises from the fact that the intrinsic quasar continuum in the Lyman-α forest region is not directly observable and is a dominant systematic. The white plot is the mean flux PCA estimated spectrum, but it clearly doesn't properly capture the intrinsic features in the forest region, after all it's a statistical fit across several spectra.
I made a crude estimation of the continuum by performing a power law fit to the region redward of the Lyα line, masking the metal emission lines and extended it upto the forest.

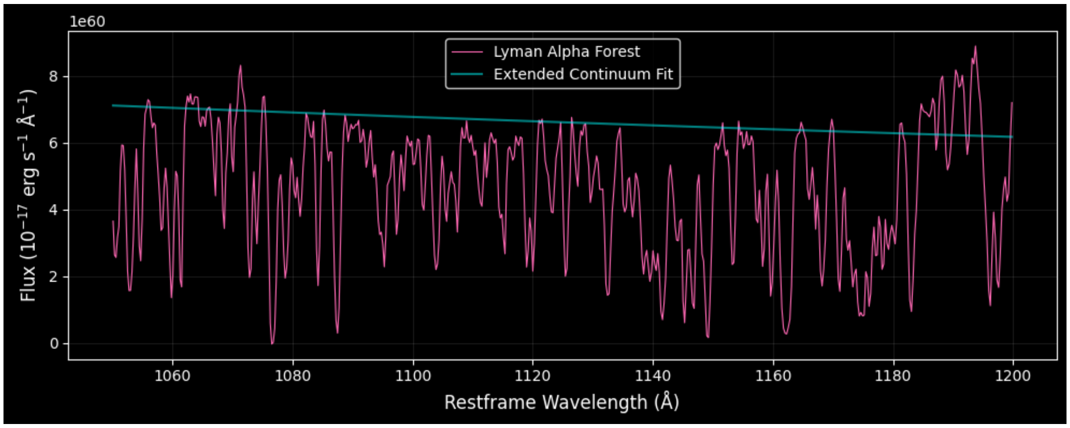
I further modelled the absorption lines as independent absorbers. The effective optical depth that provides a statistical measure of absorption across the region of the spectrum was approximated to be:

$$\tau_{\text{eff}} = -\ln\left(\frac{\sum L_{\text{obs}}}{\sum L_{\text{model}}}\right). \tag{7}$$

where the summation is over all peaks and all fluxes are calculated from the Earth frame. Using the modeled continuum, we obtain:

$$\tau_{\text{eff}} \approx 0.36. \tag{8}$$

This value is consistent with expectations for the IGM at moderate redshift according to the power law expression in [3]. Given the naive analysis, it seems that its a good order of magnitude estimate at the very least.



The extended continuum fell below the spectrum at some regions, so I selected only significant absorption troughs, setting an arbitrary filter of $\tau_i = -\ln\left(\frac{L_{\text{obs},i}}{L_{\text{cont},i}}\right) > 0.5$ for modelling each of them. Each absorption feature was modeled as an independent Gaussian in frequency space in its own rest frame:

$$\tau_i(v) = N_i \sigma_0 \frac{1}{\Delta\nu_D \sqrt{\pi}} \exp\left[-\left(\frac{\nu - \nu_0}{\Delta\nu_D}\right)^2\right]. \tag{9}$$

This approximation is valid in the regime where Doppler broadening dominates over natural (Lorentzian) broadening.

The figure clearly shows an overdensity of absorption lines close to the rest frame Lyman α frequency. The Doppler width is given by:

$$\Delta v_D = \sqrt{\frac{2kT}{m}} \frac{\nu_0}{c}. \tag{10}$$

Assuming a characteristic temperature:

$$T = 2 \times 10^4 \text{ K}, \tag{11}$$

we estimate the total column density:

$$N = \sum_i N_i \approx 2 \times 10^{15} \text{ cm}^{-2}. \tag{12}$$

This places the absorbers in the regime of typical Lyman-α forest systems rather than dense DLA systems, further justifying fitting with Gaussians.

3.2 Photoionization

The ionizing photon production rate is computed as:

$$Q = \int_{\lambda_{\min}}^{\lambda_{\max}} \frac{L_\lambda}{hc/\lambda} \, d\lambda. \tag{13}$$

With integration limits:

$$\lambda_{\min} = 820 \text{ Å}, \quad \lambda_{\max} = 912 \text{ Å}, \tag{14}$$

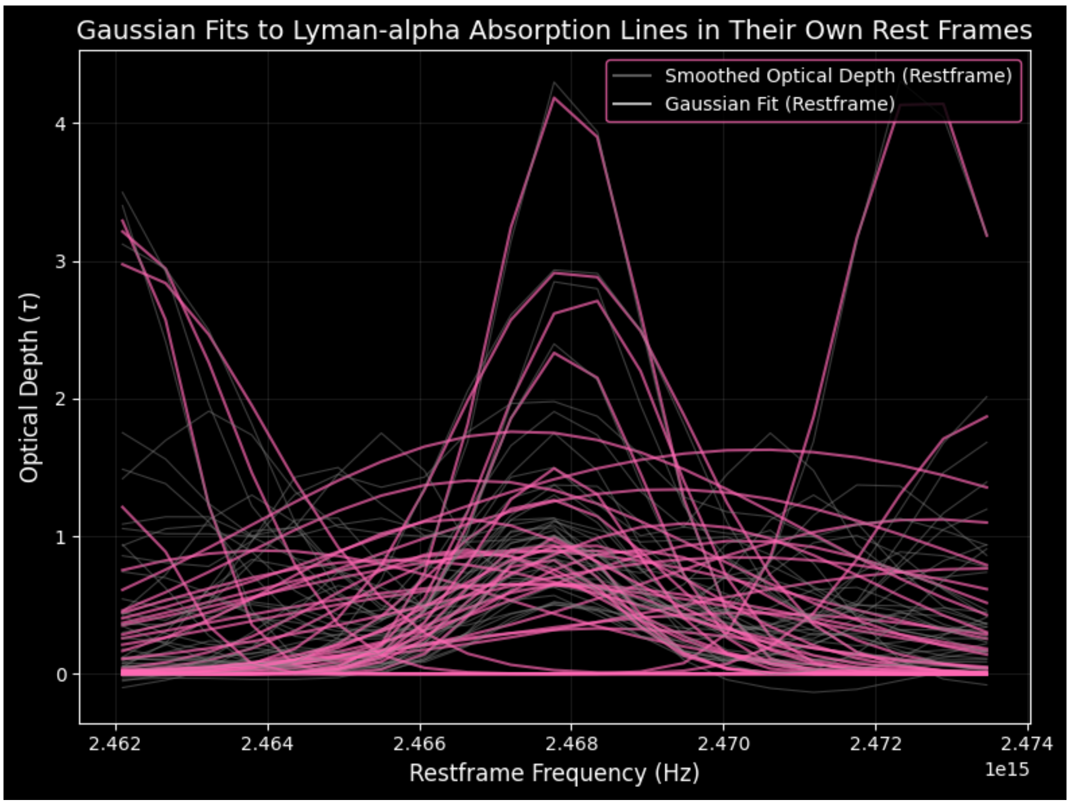


Figure 4: Plots of optical depth against frequency for each absorber in its rest frame against the best fit Gaussian.

Strictly the integration is to be done for all frequencies above the Lyman limit, however I set the lower limit to be the wavelength where all the ionising radiation is absorbed. For a more rigorous treatment, a better estimate of the continuum would be needed too.

we obtain:

$$Q \approx 3.49 \times 10^{56}\ \mathrm{s}^{-1}. \tag{15}$$

The photon flux at distance r is:

$$\Phi(r) = \frac{Q}{4\pi r^2}. \tag{16}$$

where r is the physical distance. The ionization parameter is defined as:

$$U = \frac{\Phi(r)}{n_H c}. \tag{17}$$

Approximating the hydrogen density as:

$$n_H \sim \frac{N_{\mathrm{HI}}}{\chi_{\mathrm{HI}}\, r}, \tag{18}$$

we obtain:

$$U \approx \frac{Q}{4\pi r N_{\mathrm{HI}} \chi_{\mathrm{HI}}^{-1} c}. \tag{19}$$

For representative values:

$$\chi_{\mathrm{HI}} = 10^{-5}, \quad N_{\mathrm{HI}} = 2 \times 10^{15}\ \mathrm{cm}^{-2}, \tag{20}$$

I found:

$$U \approx 0.001. \tag{21}$$

For the quasar proximity zone whose typical physical distance is 2 Mpc[2](at least to an order of magnitude), I considered:

$$\chi_{\mathrm{HI}} = 10^{-6}, \quad N_{\mathrm{HI}} = 10^{20}\ \mathrm{cm}^{-2}, \tag{22}$$

which gave

$$U \approx 10^{-6}. \tag{23}$$

3.3 Role of Thomson Scattering

To evaluate the role of scattering, I compared photon energies in hard Xray to the electron rest mass:

$$\frac{E}{m_e c^2} = \frac{100\ \mathrm{keV}}{511\ \mathrm{keV}} \approx 0.1. \tag{24}$$

which is itself much less than unity. The Lyα interaction definitely lies in the Thomson regime.

The optical depth is given by:

$$\tau_{\text{Th}} = \int n_e \sigma_T dl. \tag{25}$$

Approximating:

$$\tau_{\text{Th}} \approx N_{\text{HI}} \chi_{\text{HI}}^{-1} \sigma_T, \tag{26}$$

we obtain:

$$\tau_{\text{Th}} \sim 10^{-4}. \tag{27}$$

This is negligible compared to:

$$\tau_{\text{eff}} \approx 0.36, \tag{28}$$

demonstrating that scattering processes do not significantly contribute to the observed absorption.

4 Limitations

The results obtained in this work are broadly consistent with expectations for the Lyman-α forest. However, several important limitations must be emphasized.

1. The continuum estimation is highly uncertain and likely biases all derived quantities. For a better estimate I could use a noise model from the instrument. A good continuum model could also constrain Q better.

2. The assumption of independent Gaussian absorbers neglects line blending and LSS in the IGM. If LSS has to be taken into account it would require non-trivial calculations and one would have to do forward modelling[1].

3. The absorber temperature is degenerate with the column densities which hasn't been accounted for. For this one could couple this analysis with higher order absorption lines, essentially analyse Region B too, to break degeneracies.

4. Better constraints on the ionisation fraction would also give a better sense of U.

With these points in mind, all results have limited interpretability and are order of magnitude estimates.

5 Conclusion

I have performed a simplified analysis of the intergalactic medium using the Lyman-α forest in a quasar spectrum. Modeling absorption features, estimating column densities, and evaluating ionization conditions renders a basic picture of a highly ionized, low-density medium.

Applications of studied properties: The column densities of individual absorbers can be used to map baryon distribution along the line of sight. When LSS effects are taken into account, one can map the cosmic web using Lyα Tomography. Studying the ionization parameter gives a sense of the dominant ionization mechanism, one can constrain this transition from UV background dominance to quasar dominance by deriving U as a function of distance. If the quasar is far enough we can also study the reionization history of the universe.

To reduce assumptions, one would want to simulate more radiative transfer effects, improve continuum modelling and include more observations which would give more robust constraints.

References

[1] Benjamin Horowitz, Khee-Gan Lee, Metin Ata, Thomas Müller, Alex Krolewski, J. Xavier Prochaska, Joseph F. Hennawi, Martin White, David Schlegel, R. Michael Rich, Peter E. Nugent, Nao Suzuki, Daichi Kashino, Anton M. Koekemoer, and Brian C. Lemaux. Second data release of the cosmos ly mapping and tomography observations: The first 3d maps of the detailed cosmic web at 2.05 lt; z lt; 2.55. *The Astrophysical Journal Supplement Series*, 263(2):27, November 2022.

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[3] R. Thomas, L. Pentericci, O. Le Fèvre, A. M. Koekemoer, M. Castellano, A. Cimatti, F. Fontanot, A. Gargiulo, B. Garilli, M. Talia, R. Amorín, S. Bardelli, S. Cristiani, G. Cresci, M. Franco, J. P. U. Fynbo, N. P. Hathi, P. Hibon, Y. Khusanova, V. Le Brun, B. C. Lemaux, F. Mannucci, D. Schaerer, G. Zamorani, and E. Zucca. Less and more igm-transmitted galaxies from z 2.7 to z 6 from vandels and vuds. *Astronomy amp; Astrophysics*, 650:A63, 2021.